# Initial Exploration

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```r
#read in all data
Ellerbe_wq <- read.csv(here('Data/Raw/Ellerbe_water_quality_data.csv'),
                       stringsAsFactors = TRUE)
Sandy_wq<- read.csv(here('Data/Raw/Sandy_water_quality_data.csv'),
                       stringsAsFactors = TRUE)
```

```r
#Change Sample Date to date
Ellerbe_wq$Sample_date<-as.Date(Ellerbe_wq$Sample_date,format="%m/%d/%Y")
Sandy_wq$Sample_date<-as.Date(Sandy_wq$Sample_date,format="%m/%d/%Y")

#add a year column
Ellerbe_wq$year <- format(Ellerbe_wq$Sample_date, "%Y")
Sandy_wq$year <- format(Sandy_wq$Sample_date, "%Y")

#Data Wrangling
#dissolved oxygen
Ellerbe_wq$Dissolved_oxygen_mg.L <-
  as.numeric(as.character(Ellerbe_wq$Dissolved_oxygen_mg.L))
```

```
## Warning: NAs introduced by coercion
```

```r
Sandy_wq$Dissolved_oxygen_mg.L <-
  as.numeric(as.character(Sandy_wq$Dissolved_oxygen_mg.L))
```

```
## Warning: NAs introduced by coercion
```

```r
#Daily Mean Flow
Ellerbe_wq$Instantaneous_flow_cfs <-
  as.numeric(as.character(Ellerbe_wq$Instantaneous_flow_cfs))
```

```
## Warning: NAs introduced by coercion
```

```r
Sandy_wq$Instantaneous_flow_cfs <-
  as.numeric(as.character(Sandy_wq$Instantaneous_flow_cfs))
```

```
## Warning: NAs introduced by coercion
```

```r
#Temperature
Ellerbe_wq$Water_temperature_C <-
  as.numeric(as.character(Ellerbe_wq$Water_temperature_C))
```

```
## Warning: NAs introduced by coercion
```

```r
Sandy_wq$Water_temperature_C <-
  as.numeric(as.character(Sandy_wq$Water_temperature_C))
```

```
## Warning: NAs introduced by coercion
```

```r
#removing weird symbols and converting to numeric
Ellerbe_wq <- Ellerbe_wq %>%
  mutate(
    Phosphorus_unfiltered_mg.L_P = as.character(Phosphorus_unfiltered_mg.L_P),
    Phosphorus_unfiltered_mg.L_P = str_remove_all(Phosphorus_unfiltered_mg.L_P, "<"),
    Phosphorus_unfiltered_mg.L_P = if_else(
      str_detect(Phosphorus_unfiltered_mg.L_P, "E"),
      NA_character_,
      Phosphorus_unfiltered_mg.L_P
    ),
    Phosphorus_unfiltered_mg.L_P = as.numeric(Phosphorus_unfiltered_mg.L_P)
  )

Sandy_wq <- Sandy_wq %>%
  mutate(
    Phosphorus_unfiltered_mg.L_P = as.character(Phosphorus_unfiltered_mg.L_P),
    Phosphorus_unfiltered_mg.L_P = str_remove_all(Phosphorus_unfiltered_mg.L_P, "<"),
    Phosphorus_unfiltered_mg.L_P = if_else(
      str_detect(Phosphorus_unfiltered_mg.L_P, "E"),
      NA_character_,
      Phosphorus_unfiltered_mg.L_P
    ),
    Phosphorus_unfiltered_mg.L_P = as.numeric(Phosphorus_unfiltered_mg.L_P)
  )
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Phosphorus_unfiltered_mg.L_P =
##   as.numeric(Phosphorus_unfiltered_mg.L_P)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```r
Ellerbe_wq <- Ellerbe_wq %>%
  mutate(
    Total.nitrogen_computed_mg.L_N = as.character(Total.nitrogen_computed_mg.L_N),
    Total.nitrogen_computed_mg.L_N = str_remove_all(Total.nitrogen_computed_mg.L_N, "<"),
    Total.nitrogen_computed_mg.L_N = if_else(
      str_detect(Total.nitrogen_computed_mg.L_N, "E"),
      NA_character_,
      Total.nitrogen_computed_mg.L_N
    ),
    Total.nitrogen_computed_mg.L_N = as.numeric(Total.nitrogen_computed_mg.L_N)
  )
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Total.nitrogen_computed_mg.L_N =
##   as.numeric(Total.nitrogen_computed_mg.L_N)`.
## Caused by warning:
## ! NAs introduced by coercion
```

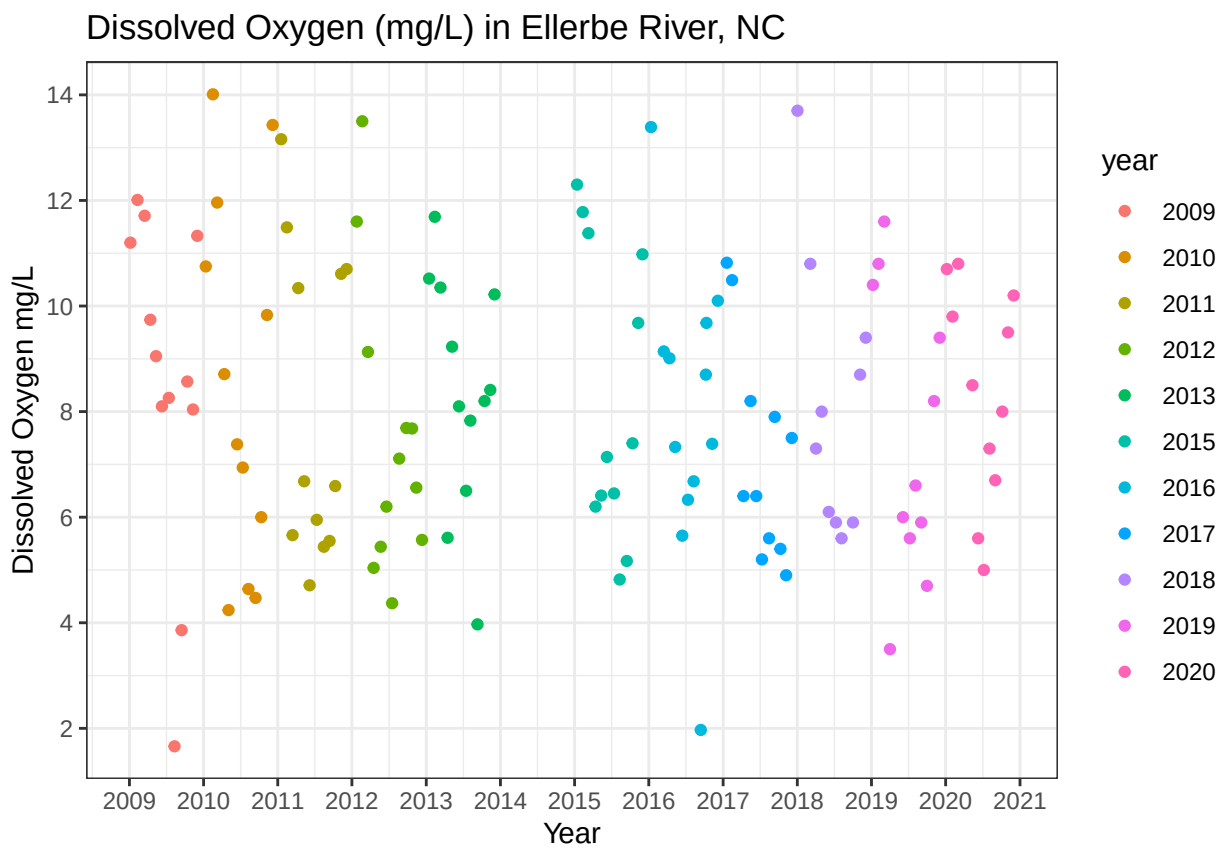
```
Sandy_wq <- Sandy_wq %>%
  mutate(
    Total.nitrogen_computed_mg.L_N = as.character(Total.nitrogen_computed_mg.L_N),
    Total.nitrogen_computed_mg.L_N = str_remove_all(Total.nitrogen_computed_mg.L_N, "<"),
    Total.nitrogen_computed_mg.L_N = if_else(
      str_detect(Total.nitrogen_computed_mg.L_N, "E"),
      NA_character_,
      Total.nitrogen_computed_mg.L_N
    ),
    Total.nitrogen_computed_mg.L_N = as.numeric(Total.nitrogen_computed_mg.L_N)
  )
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Total.nitrogen_computed_mg.L_N =
##   as.numeric(Total.nitrogen_computed_mg.L_N)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```
#removing useless columns
Sandy_wq <- Sandy_wq[, -c(18, 19,20)]
Ellerbe_wq <- Ellerbe_wq[, -c(17, 18)]
```

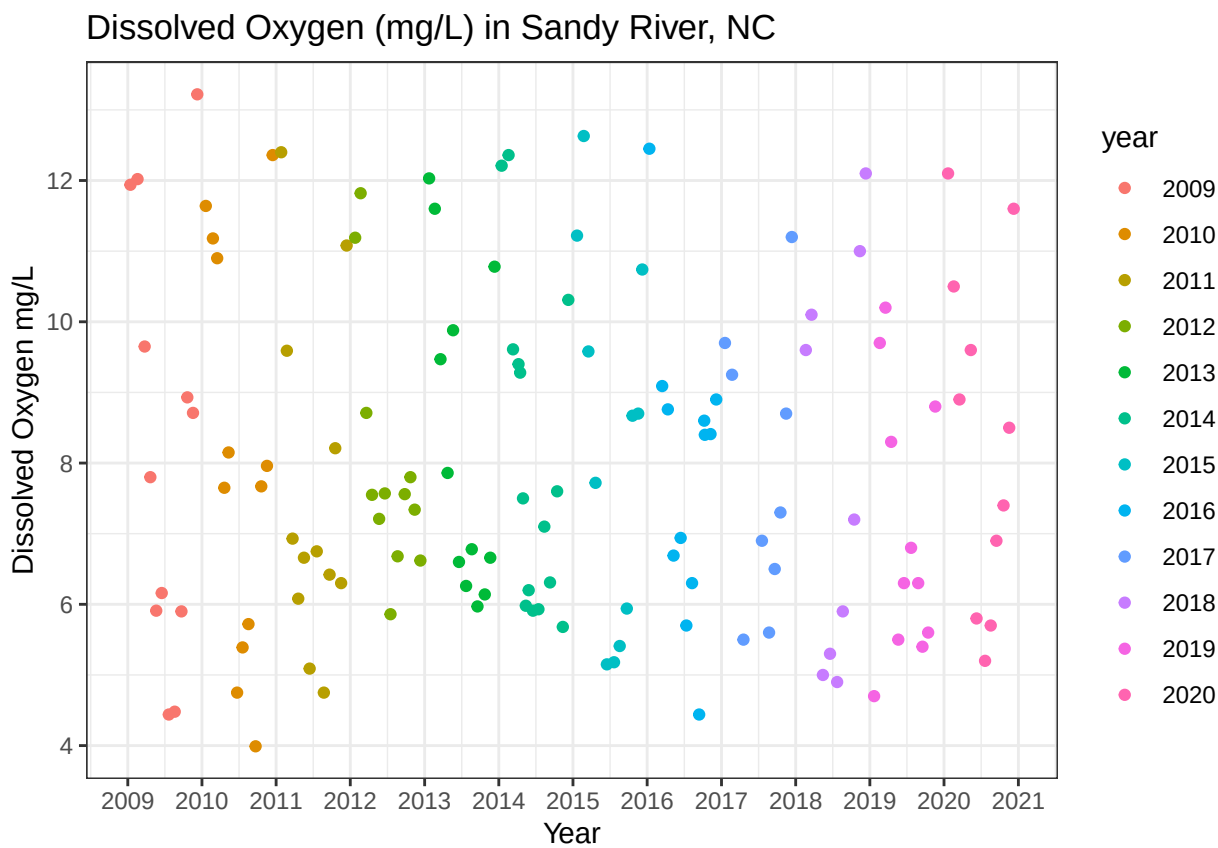
```
#Plot Dissolved Oxygen in Ellerbe over the years
ggplot(Ellerbe_wq, aes(x=Sample_date,
                       y=Dissolved_oxygen_mg.L, color=year))+
  geom_point()+
   labs(title = "Dissolved Oxygen (mg/L) in Ellerbe River, NC",
        x = "Year",
y = "Dissolved Oxygen mg/L")+
   scale_y_continuous(breaks = c(2.0, 4.0, 6.0, 8.0, 10.0, 12.0, 14.0))+
   scale_x_date( breaks = date_breaks("1 year"),
                 labels = date_format("%Y"))+
  theme_bw()
```

```
## Warning: Removed 26 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
#Plot Dissolved Oxygen in Sandy over the years
ggplot(Sandy_wq, aes(x=Sample_date,
                     y=Dissolved_oxygen_mg.L, color=year))+
  geom_point()+
   labs(title = "Dissolved Oxygen (mg/L) in Sandy River, NC",
            x = "Year",
y = "Dissolved Oxygen mg/L")+
   scale_y_continuous(breaks = c(2.0, 4.0, 6.0, 8.0, 10.0, 12.0, 14.0))+
   scale_x_date( breaks = date_breaks("1 year"),
                 labels = date_format("%Y"))+
  theme_bw()
```

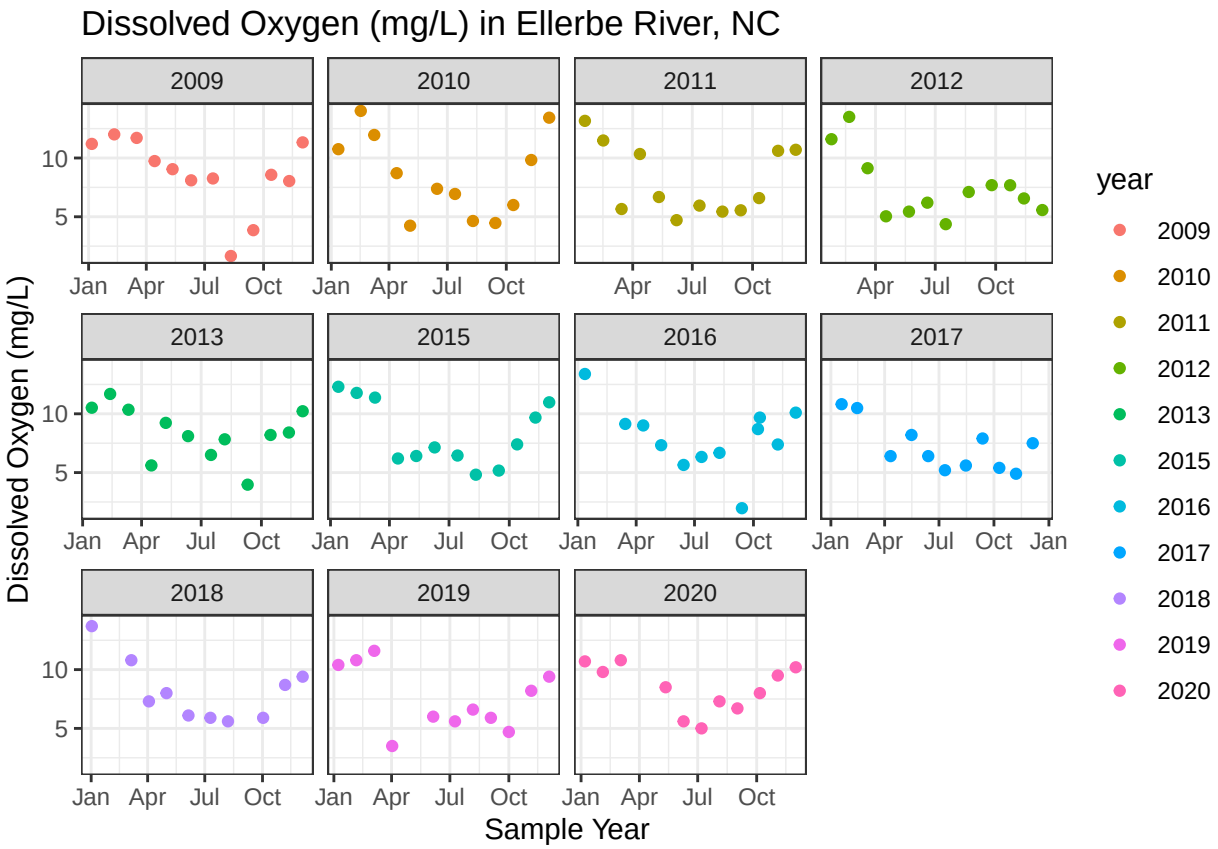
```
## Warning: Removed 29 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




#Dissolved Oxygen by River

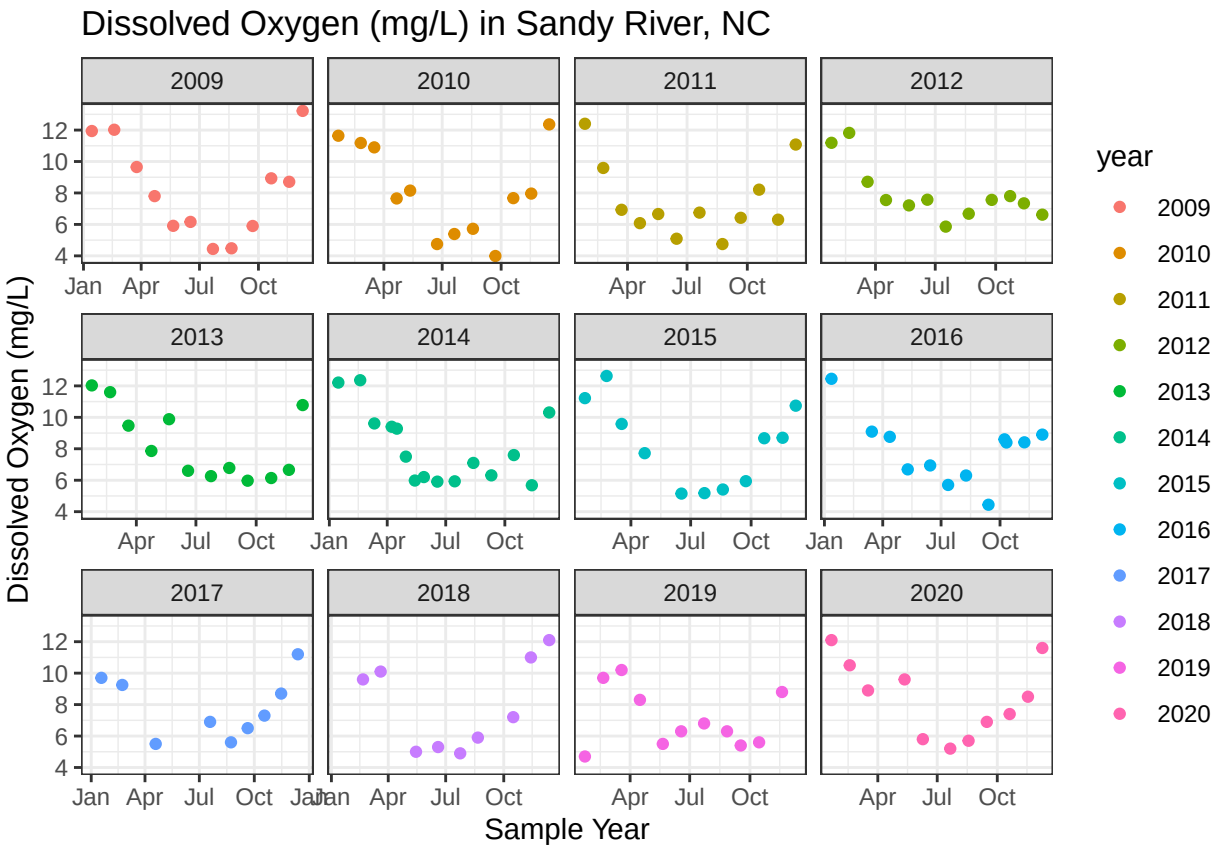
```
#Plot Dissolved Oxygen in Ellerbe each year facet wrapped
ggplot(Ellerbe_wq, aes(x=Sample_date,
                       y=Dissolved_oxygen_mg.L, color=year))+
  geom_point()+
   labs(title = "Dissolved Oxygen (mg/L) in Ellerbe River, NC",
        x = "Sample Year",
y = "Dissolved Oxygen (mg/L)")+
   facet_wrap(~year, scales = "free_x")+
  scale_x_date(date_labels = "%b") +
  theme_bw()
```

```
## Warning: Removed 26 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
#Plot Dissolved Oxygen in Sandy each year facet wrapped
ggplot(Sandy_wq, aes(x=Sample_date,
                     y=Dissolved_oxygen_mg.L, color=year))+
  geom_point()+
   labs(title = "Dissolved Oxygen (mg/L) in Sandy River, NC",
        x = "Sample Year",
y = "Dissolved Oxygen (mg/L)")+
   facet_wrap(~year, scales = "free_x")+
    scale_x_date(date_labels = "%b") +
  theme_bw()
```

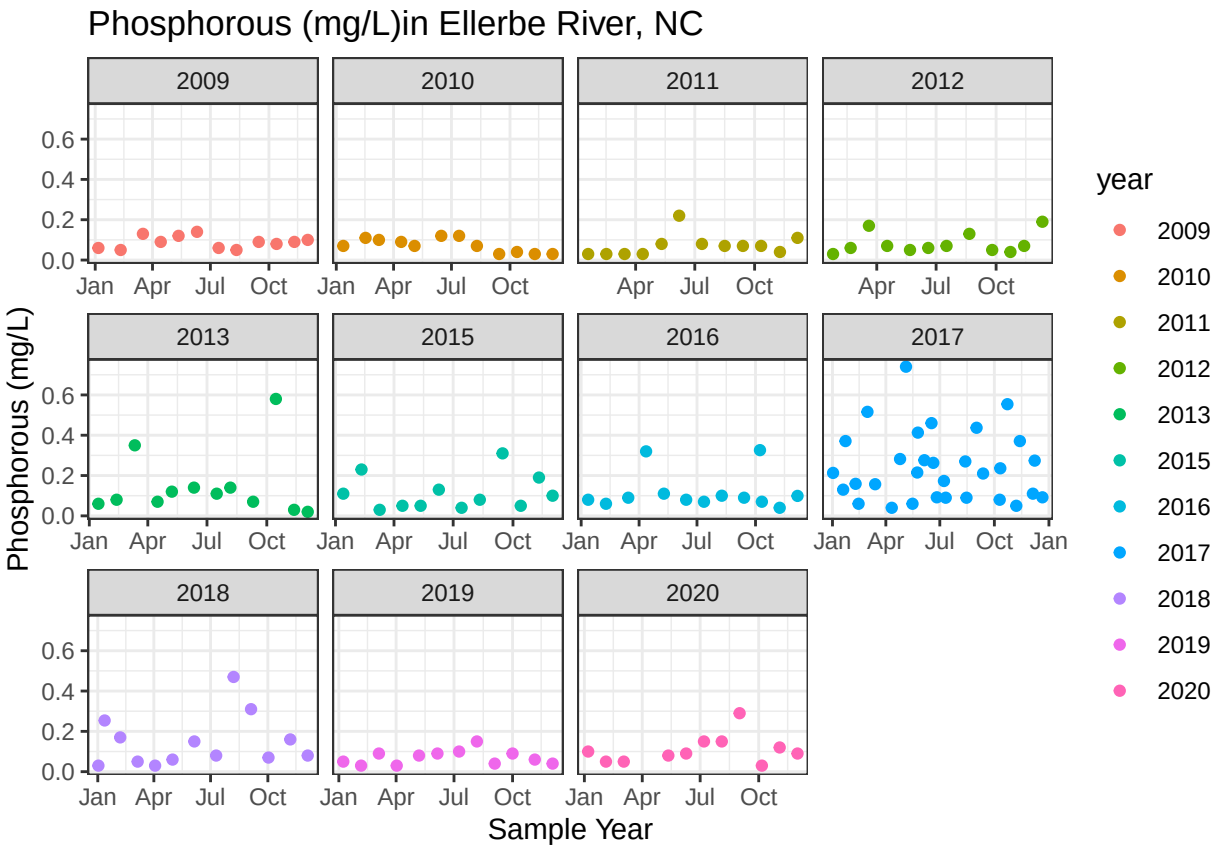
```
## Warning: Removed 29 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




#Phosphorous by River

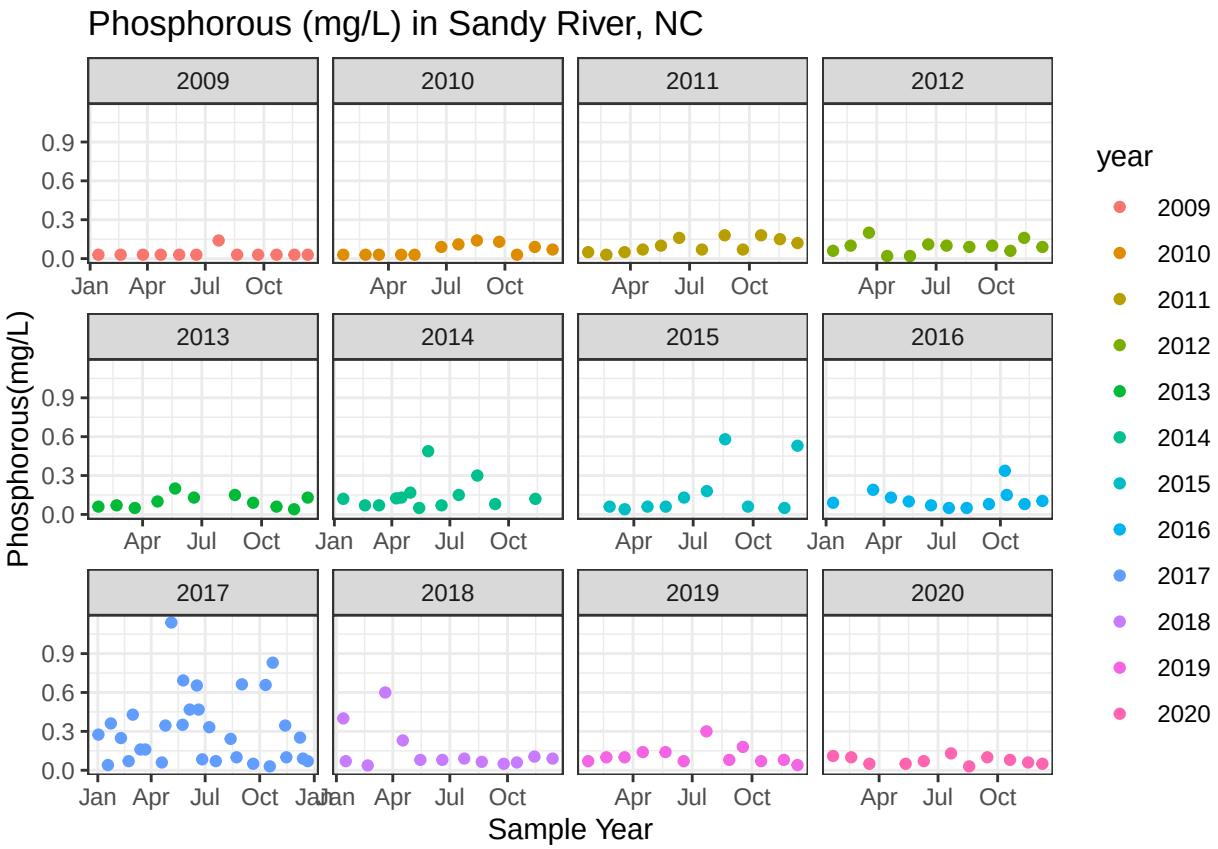
```
#Plot Phosphorous in Ellerbe each year
ggplot(Ellerbe_wq, aes(x=Sample_date,
                       y=Phosphorus_unfiltered_mg.L_P, color=year))+
  geom_point()+
   labs(title = "Phosphorous (mg/L)in Ellerbe River, NC",
        x = "Sample Year",
y = "Phosphorous (mg/L)")+
   facet_wrap(~year, scales = "free_x")+
    scale_x_date(date_labels = "%b") +
  theme_bw()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
#Plot Phosphorous in Sandy each year
ggplot(Sandy_wq, aes(x=Sample_date,
                     y=Phosphorus_unfiltered_mg.L_P, color=year))+
  geom_point()+
   labs(title = "Phosphorous (mg/L) in Sandy River, NC",
        x = "Sample Year",
y = "Phosphorous(mg/L)")+
   facet_wrap(~year, scales = "free_x")+
    scale_x_date(date_labels = "%b") +
  theme_bw()
```

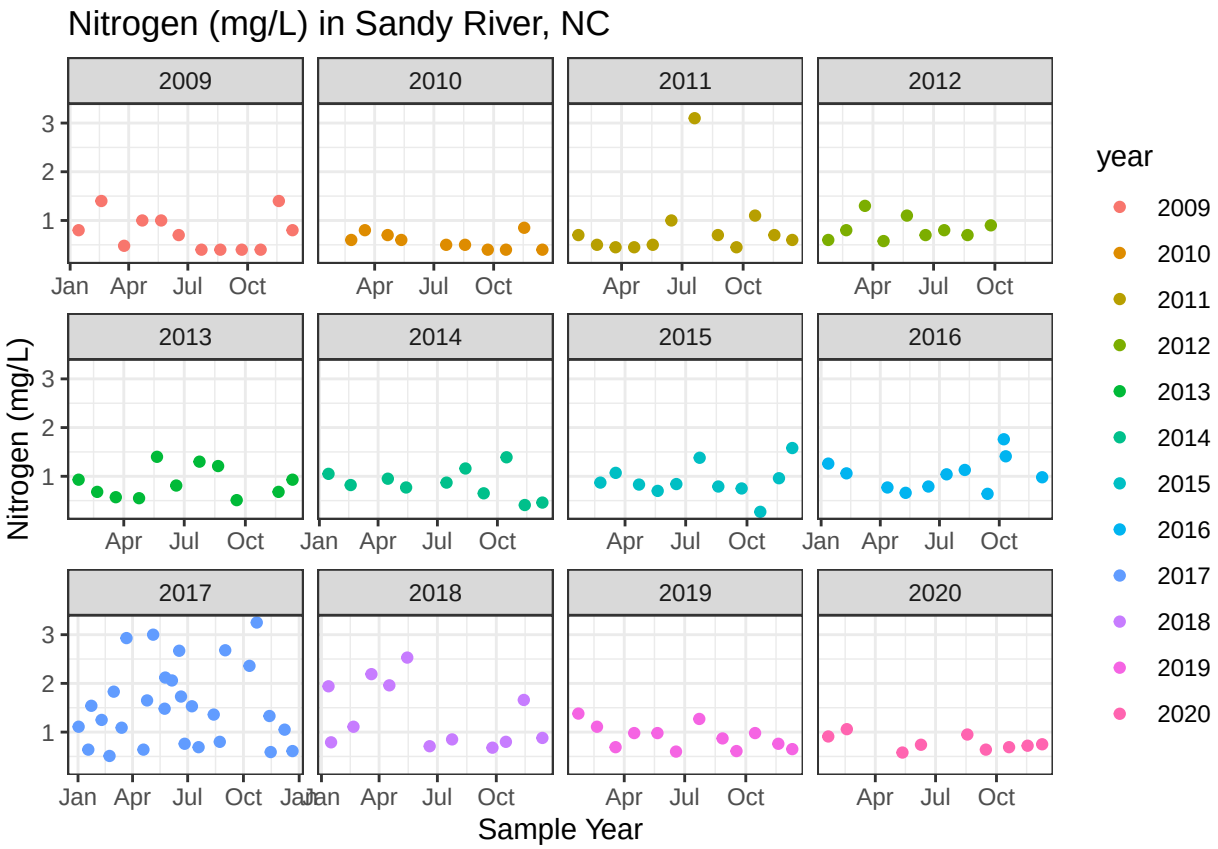
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




#Nitrogen

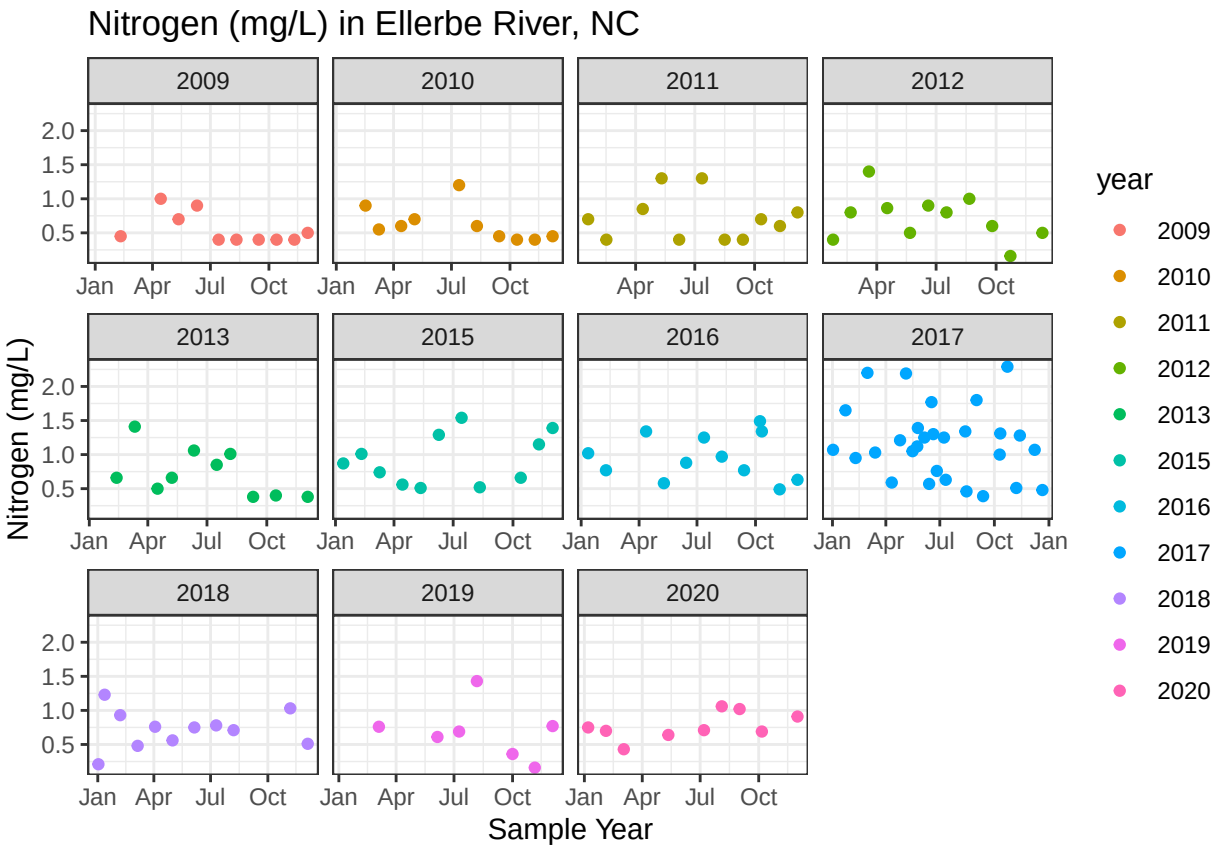
```
#Plot Nitrogen in Sandy each year
ggplot(Sandy_wq, aes(x=Sample_date,
                     y=Total.nitrogen_computed_mg.L_N, color=year))+
  geom_point()+
   labs(title = "Nitrogen (mg/L) in Sandy River, NC",
        x = "Sample Year",
y = "Nitrogen (mg/L)")+
   facet_wrap(~year, scales = "free_x")+
    scale_x_date(date_labels = "%b") +
  theme_bw()
```

```
## Warning: Removed 20 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
#Plot Nitrogen in Ellerbe each year
ggplot(Ellerbe_wq, aes(x=Sample_date,
                       y=Total.nitrogen_computed_mg.L_N, color=year))+
  geom_point()+
   labs(title = "Nitrogen (mg/L) in Ellerbe River, NC",
        x = "Sample Year",
y = "Nitrogen (mg/L)")+
   facet_wrap(~year, scales = "free_x")+
    scale_x_date(date_labels = "%b") +
  theme_bw()
```

```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# Export
write.csv(Ellerbe_wq, here("Data/Processed/Ellerbe_wq_clean.csv"), row.names = FALSE)

write.csv(Sandy_wq, here("Data/Processed/Sandy_wq_clean.csv"), row.names = FALSE)
```